Pneumonia Navigator

Participant workbook and decision log

Module F · Participant guide

Synthetic teaching data

Instats · August 5, 2026

The facilitator operates the Navigator. You co-author the audit.

1. Make a decision before the prepared entry appears.
2. Record the evidence that supports your choice.
3. Name what would make you revise it.
4. Track the downstream consequence of that revision.

Your deliverable

One Navigator entry you can accept, correct, or reject with an evidence-based explanation.

Manual route—no login required

`https://navigator.tao-rwd.com/`

Export the completed record as Markdown at Step 8.

1. Start with the causal question

Among adults in a synthetic health-system cohort, what is the effect of pneumococcal vaccination at baseline, versus no vaccination, on 12-month pneumonia hospitalization?

Observed association

Vaccinated risk: **7.19%** Unvaccinated risk: **6.43%**

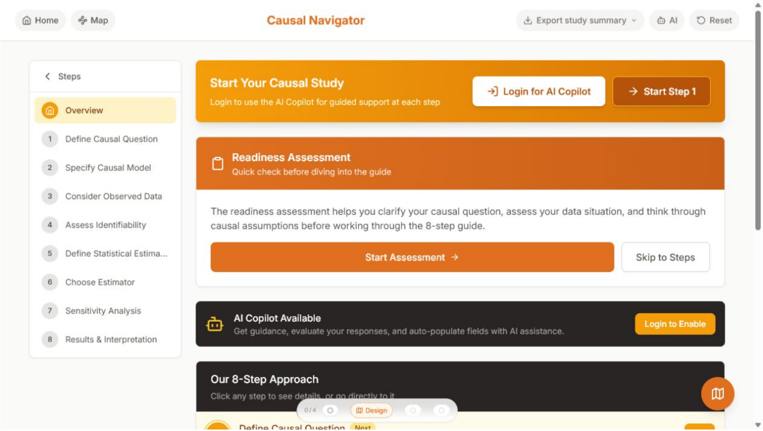
Crude risk difference: $+0.76$ **percentage points**

Before interpreting it

Write the population, treatment, comparator, time zero, outcome window, and causal contrast you need.

Synthetic teaching result — not clinical evidence.

The Navigator is an eight-step spine



1. Question
2. Causal model
3. Observed data
4. Identification
5. Estimand
6. Estimator
7. Sensitivity
8. Interpretation

Each answer constrains the next.

Checkpoint 1: specify Step 1 before the reveal

Population

Treatment

Comparator

Time zero

Outcome window

Causal contrast

Revision trigger

Which single phrase in your specification would most change the target estimand?

Checkpoint 2: map model to data

STIPULATED BASELINE NODES

- age
- prior pneumonia
- prior vaccination
- current vaccination A
- pneumonia hospitalization Y

Model decision

Draw or describe the arrows. Which baseline variables belong in the adjustment set, and why?

OBSERVED DATA

People	5,000
Vaccinated	1,627
Pneumonia events	334
Missingness	none
Censoring	none in the fixed data

Mapping decision

Can every Step 1 element be represented by a variable and measurement window in these data?

Checkpoint 3: audit identification

Assumption	Evidence you would seek	What would make you revise it
Exchangeability		
Positivity		
Consistency		
No interference		

Choose one

Which assumption is weakest in real EHR data? The fixed dataset cannot certify these scientific assumptions.

Checkpoint 4: target before method

CAUSAL TARGET

$$E[Y^1] - E[Y^0]$$

STATISTICAL ESTIMAND

$$E_W\left[E(Y \mid A = 1, W)\right.$$
$$\left. - E(Y \mid A = 0, W)\right]$$

Link

Which Step 4 assumptions make these quantities coincide?

PROPOSED ESTIMATOR

TMLE

- $\bar{Q}(A, W)$: outcome regression
- $g(W)$: treatment mechanism
- influence-function interval

Boundary

Name one problem TMLE can reduce and one problem it cannot repair.

Checkpoint 4 continued: plan how the conclusion could fail

Unmeasured confounding	Could residual frailty or healthcare-seeking change the direction?
Outcome error	Would imperfect pneumonia coding change the direction or magnitude?
Selection	What if death, disenrollment, or EHR capture is informative?
Estimation	Do alternative learners or propensity truncation change the result?

Choose one gap

Specify one sensitivity analysis or one new source of evidence. A confidence interval does not measure these causal gaps.

Checkpoint 5: disciplined conclusion

PROPOSED STATEMENT A

“In this synthetic exercise, adjustment reverses the crude association under the stipulated causal model.”

PROPOSED STATEMENT B

“The vaccine prevents pneumonia in real patients.”

Accept, correct, or reject

For each statement, cite one assumption or limitation. Rewrite the weaker statement so it says no more than the design supports.

Your turn: defend one Navigator entry

Choose one field from Steps 1–8.

1. What evidence makes the entry defensible?
2. What would make you revise it?
3. Which downstream decision changes if you revise it?

Use the answer key after the exercise

Compare your reasoning with `causal-roadmap-pneumonia-worked-example.md`; do not substitute it for your argument.